Intelligent Enhanced Oil Recovery Method Selection Using Random Forest and Artificial Neural Network

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Abstract

Enhanced Oil Recovery (EOR) refers to techniques used to increase the amount of crude oil that can be extracted from an oil reservoir beyond what is achievable through primary and secondary recovery. Selecting the appropriate EOR method for a given reservoir is a critical and complex decision that depends on a multitude of reservoir properties. This paper presents an intelligent, data-driven framework for EOR method selection employing two complementary machine learning approaches: (1) a Random Forest (RF) classifier that leverages ensemble decision trees for robust, interpretable classification across four EOR categories—thermal, chemical, gas injection, and miscible flooding—and (2) an Artificial Neural Network (ANN) that models non-linear feature-label relationships for higher classification accuracy. Both models are trained on a curated reservoir dataset comprising features such as oil viscosity, reservoir depth, permeability, porosity, temperature, and API gravity. The RF model achieves a classification accuracy of 91.4%, while the ANN reaches 94.7% on the held-out test set. Feature importance analysis from the RF highlights oil viscosity and API gravity as the most discriminating reservoir properties. The proposed framework provides field engineers with a reliable decision-support tool, reducing reliance on manual screening charts and expert judgment alone.

Index Terms—Enhanced Oil Recovery; Random Forest; Artificial Neural Network; Reservoir characterization; Machine learning; Classification; Decision support; Petroleum engineering.

I. INTRODUCTION

Global energy demand continues to rise while the discovery of new large-scale oil reserves has significantly declined. As a result, the extraction efficiency of existing reservoirs has become a priority for the petroleum industry. Primary recovery—relying solely on natural reservoir pressure—typically yields only 5–15% of the original oil in place (OOIP), while secondary recovery methods such as waterflooding can raise cumulative recovery to 25–35% [1]. Enhanced Oil Recovery (EOR) techniques are applied as tertiary methods to mobilize the remaining trapped hydrocarbons, with the potential to recover an additional 5–20% of OOIP depending on the reservoir and the method employed.

EOR methods are broadly categorized into four families:

- Thermal methods: Steam flooding, cyclic steam stimulation, and in-situ combustion—primarily suited for heavy crude oils with high viscosity.
- Chemical methods: Surfactant, polymer, and alkaline flooding—effective in moderate-viscosity reservoirs with low-to-moderate salinity.
- Gas injection: CO_2, nitrogen, or hydrocarbon gas injection—commonly applied to light oil reservoirs at adequate depths and pressures.
- Miscible flooding: A subset of gas injection where injected gas achieves miscibility with reservoir oil, dramatically reducing interfacial tension.

Selecting the optimal EOR method requires simultaneous evaluation of numerous reservoir properties: oil gravity (API), viscosity, depth, temperature, permeability, porosity, formation water salinity, and net pay thickness. Traditional EOR screening relies on empirical screening guidelines developed by Taber, Martin, and Seright [2], presented as parameter ranges for each method. While these charts are widely used, they provide only binary yes/no decisions per method and do not rank or prioritize options when multiple methods qualify. Machine learning offers a promising pathway to move beyond static screening criteria by learning decision boundaries directly from historical EOR project data. In this paper, we present and compare two ML-based classification models—Random Forest and Artificial Neural Network—for automated

EOR method recommendation. The key contributions of this work are:

1) A curated, feature-engineered EOR reservoir dataset with four target classes.
2) A comparative study of RF and ANN for multi-class EOR classification.
3) Feature importance analysis to identify the most discriminating reservoir parameters.
4) A decision-support framework deployable in field engineering workflows.

II. RELATED WORK

Early EOR screening was conducted using parameter range tables compiled from field experience [2]. These tables define minimum and maximum values for reservoir properties acceptable for each EOR method and require the engineer to manually compare reservoir characteristics against each range. With the advent of data-driven methods, several researchers have explored machine learning for EOR selection and performance prediction. Surguchev et al. applied fuzzy logic to match reservoir parameters to EOR categories, improving upon binary threshold screening. Hou et al. employed support vector machines (SVM) for EOR method classification and demonstrated improved accuracy over classical screening charts for a dataset of 100 reservoirs [3]. Choubineh et al. used an adaptive neuro-fuzzy inference system (ANFIS) to predict oil recovery factors under polymer flooding, achieving satisfactory predictive performance [4].

More recently, ensemble learning methods have been explored for reservoir property prediction. Anifowose et al. demonstrated that ensemble approaches outperform single classifiers for permeability prediction in heterogeneous reservoirs [5]. Deep learning architectures have also been applied to well-log interpretation and reservoir characterization.

However, a direct, systematic comparison between Random Forest and ANN specifically for multi-class EOR method selection, accompanied by feature importance attribution, remains limited in the literature. This work addresses that gap by providing a structured benchmarking study with engineeringinterpretable outputs.

III. DATASET DESCRIPTION

A. Data Source and Composition

The dataset used in this study is compiled from published EOR project records, reservoir engineering literature, and open petroleum databases. Each record corresponds to a single reservoir segment for which an EOR method has been historically applied and documented. After data cleaning and removal of records with excessive missing values, the final dataset comprises 850 reservoir samples distributed across four EOR classes.

B. Feature Set

Seven reservoir and fluid properties are selected as input features, based on their prominence in EOR screening guidelines and their measurability in standard reservoir characterization workflows. Table I summarizes the feature set.

TABLE I
INPUT FEATURE DESCRIPTIONS

Feature	Unit	Description
API Gravity	°API	Oil density measure
Oil Viscosity	cp	Resistance to flow
Reservoir Depth	ft	Depth to pay zone
Reservoir Temp.	°F	Formation temperature
Permeability	mD	Flow capacity
Porosity	fraction	Void space ratio
Net Pay Thickness	ft	Productive zone height

C. Target Classes

The target variable is the recommended EOR category, encoded as follows:

- Class 0 — Thermal EOR
- Class 1 — Chemical EOR
- Class 2 — Gas Injection
- Class 3 — Miscible Flooding

D. Class Distribution

Table II presents the distribution of samples across the four EOR categories in the dataset.

TABLE II
DATASET CLASS DISTRIBUTION

EOR Method	Samples	Percentage
Thermal EOR	248	29.2%
Chemical EOR	196	23.1%
Gas Injection	231	27.2%
Miscible Flooding	175	20.6%
Total	850	100%

IV. METHODOLOGY

A. Data Preprocessing

Prior to model training, the following preprocessing steps are applied:

1) Missing value imputation: Features with fewer than 5% missing values are imputed using the column median. Records with more than 30% missing features are removed.
2) Outlier handling: Interquartile range (IQR) capping is applied to clip extreme values at the 1st and 99th percentiles.

3) Feature scaling: Min-Max normalization is applied to all features to bring them to the range $[0,1]$, which is particularly important for the ANN model.
4) Train/test split: The dataset is split into 80% training (680 samples) and 20% testing (170 samples) using stratified sampling to preserve class proportions.

B. Random Forest Classifier

Random Forest is an ensemble learning method that constructs a large number of decision trees during training and outputs the class that is the mode of the individual tree predictions [6]. For a dataset with N samples and M features, each tree is trained on a bootstrap sample of N records. At each node split, a random subset of m features ($m \ll M$) is considered, reducing correlation between trees and improving generalization.

The prediction for a new input $\mathbf{x}$ is:

$$\hat{y} = \text{mode}\{h_t(\mathbf{x})\}_{t=1}^{T} \tag{1}$$

where $h_t(\mathbf{x})$ is the prediction of the t-th tree and T is the total number of trees.

Feature importance in RF is computed as the mean decrease in Gini impurity across all trees when a feature is used for splitting. The Gini impurity of a node n is:

$$G(n) = 1 - \sum_{k=1}^{K} p_k^2 \tag{2}$$

where p_k is the proportion of class k samples at node n.

1) Hyperparameter Configuration: Table III lists the final hyperparameter configuration for the RF model, determined via 5-fold cross-validation grid search.

TABLE III RANDOM FOREST HYPERPARAMETERS

Parameter	Value
Number of trees (T)	200
Max features per split (m)	$\sqrt{M}$
Max tree depth	15
Min samples per leaf	3
Bootstrap sampling	Enabled
Class weight	Balanced
Random state	42

C. Artificial Neural Network

The ANN is a fully connected feedforward network trained to classify reservoir samples into one of the four EOR categories. The network maps the 7-dimensional input feature vector through two hidden layers to a 4-class softmax output. The architecture is summarized in Table IV.

TABLE IV
ANN ARCHITECTURE SUMMARY

Layer	Type	Units	Activation
Input	Dense	7	—
Hidden 1	Dense	64	ReLU
Dropout	—	—	(0.3)
Hidden 2	Dense	32	ReLU
Dropout	—	—	(0.2)
Output	Dense	4	Softmax

The network is trained using the Adam optimizer with a learning rate of $\alpha = 0.001$ and categorical cross-entropy loss:

$$L = -\sum_{i=1}^{N}\sum_{k=1}^{K} y_{ik} \log(\hat{y}_{ik}) \tag{3}$$

where y_{ik} is the one-hot encoded ground truth and $\hat{y}_{ik}$ is the predicted probability for class k of sample i.

Training is conducted for 150 epochs with a batch size of 32. Early stopping with a patience of 20 epochs monitors validation loss to prevent overfitting.

D. Evaluation Metrics

Both models are evaluated using the following metrics on the held-out test set:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{4}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{5}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{6}$$

$$F_1 = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{7}$$

V. EXPERIMENTAL RESULTS

A. Random Forest Performance

The RF model achieves an overall test accuracy of 91.4%. Table V presents the per-class precision, recall, and F1-score.

TABLE V
RANDOM FOREST CLASSIFICATION REPORT

EOR Class	Precision	Recall	F1	Support
Thermal EOR	0.94	0.93	0.94	50
Chemical EOR	0.88	0.90	0.89	39
Gas Injection	0.91	0.92	0.91	46
Miscible Flooding	0.91	0.89	0.90	35
Macro avg	0.91	0.91	0.91	170

B. Feature Importance (Random Forest)

The RF model provides interpretable feature importance scores via mean Gini impurity decrease. Table VI ranks the seven input features by their importance scores.

TABLE VI
FEATURE IMPORTANCE SCORES (RANDOM FOREST)

Feature	Importance Score	Rank
Oil Viscosity	0.248	1
API Gravity	0.221	2
Reservoir Depth	0.183	3
Reservoir Temp.	0.142	4
Permeability	0.103	5
Porosity	0.063	6
Net Pay Thickness	0.040	7

Oil viscosity and API gravity together account for nearly 47% of the model's discriminative power. This aligns with petroleum engineering intuition: thermal methods target highviscosity, low-API crudes, while gas injection and miscible flooding are suited to light oil (high API).

C. ANN Performance

The ANN achieves an overall test accuracy of 94.7%, outperforming the RF by 3.3 percentage points. Table VII presents per-class performance metrics.

The ANN benefits from its ability to model non-linear decision boundaries across the 7-dimensional feature space, which may explain the performance gain over the RF, particularly for Chemical EOR and Miscible Flooding classes that share overlapping parameter ranges.

TABLE VII
ANN CLASSIFICATION REPORT

EOR Class	Precision	Recall	F1	Support
Thermal EOR	0.96	0.96	0.96	50
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Miscible Flooding	0.94	0.94	0.94	35
Macro avg	0.94	0.95	0.94	170

D. Training Dynamics (ANN)

Training and validation accuracy were monitored across 150 epochs. Training accuracy reached 97.2% by epoch 80, after which early stopping was not triggered, indicating stable convergence. Validation loss plateaued at approximately epoch 110 without divergence, confirming that the dropout layers effectively regularized the network.

E. Comparative Analysis

Table VIII provides a side-by-side comparison of both approaches across key dimensions.

TABLE VIII
COMPARISON OF RF AND ANN APPROACHES

Attribute	Random Forest	ANN
Test Accuracy	91.4%	94.7%
Training Time	~12 s	~45 s
Interpretability	High (FI)	Low (Black-box)
Hyperparameter Tuning	Moderate	High
Overfitting Risk	Low	Moderate
Feature Importance	Native	Requires SHAP
Inference Speed	Fast	Fast
Deployment Size	Small	Moderate

VI. SYSTEM DESIGN AND PIPELINE

The end-to-end EOR selection framework consists of four modules:

1) Data Ingestion Module: Accepts reservoir characterization data via CSV upload or manual parameter entry through a web interface built with Streamlit.
2) Preprocessing Module: Applies imputation, outlier capping, and normalization as described in Section IV-A.
3) Model Inference Module: Passes preprocessed features to both the trained RF and ANN models, generating class probability distributions.
4) Decision Output Module: Displays the recommended EOR method, class confidence scores, and the RF feature importance chart to support interpretability for field engineers.

Both models are serialized using Scikit-learn's joblib for the RF and TensorFlow SavedModel format for the ANN, enabling efficient reloading without retraining.

VII. APPLICATIONS AND ADVANTAGES

The proposed EOR method selection system offers value across multiple domains in the petroleum industry:

- Field Screening: Automates preliminary EOR screening for large portfolios of well candidates, reducing time from weeks to seconds.
- Decision Support: Provides probabilistic recommendations alongside feature importance, enabling engineers to validate model outputs against domain knowledge.
- Asset Optimization: Supports reservoir asset teams in prioritizing EOR investments by ranking multiple reservoir candidates against each other.
- Education and Training: Serves as a teaching tool for petroleum engineering students to understand the link between reservoir properties and EOR suitability.

- Edge Deployment: The RF model's lightweight footprint makes it suitable for deployment on tablets or offline field computing environments.

VIII. FUTURE ENHANCEMENTS

Several directions are identified for extending this work:

- Expanded Dataset: Incorporating additional EOR project records from public databases (e.g., SPE data repositories) to improve model generalization.
- Additional EOR Categories: Extending classification to include SAGD, VAPEX, and microbial EOR methods currently absent from the label space.
- SHAP Explainability: Integrating SHAP (SHapley Additive exPlanations) values for the ANN to provide perprediction feature attribution, improving trust among petroleum engineers.
- Economic Integration: Incorporating capital expenditure and operating cost parameters to enable costeffectiveness ranking alongside technical suitability.
- Time-Series Prediction: Extending the framework to predict EOR recovery factor trajectories over time using LSTM networks.
- Web Application: Developing a cloud-hosted web application with REST API for integration with commercial reservoir simulation software.

IX. CONCLUSION

This paper presented an intelligent data-driven framework for Enhanced Oil Recovery method selection using Random Forest and Artificial Neural Network classifiers. Both models were trained on a curated reservoir dataset of 850 samples spanning four EOR categories: thermal, chemical, gas injection, and miscible flooding. The Random Forest achieved 91.4% test accuracy and provided interpretable feature importance rankings that confirmed the primacy of oil viscosity and API gravity in EOR screening. The ANN achieved 94.7% accuracy by leveraging its capacity to model non-linear feature interactions. Together, the two approaches offer complementary strengths: the RF for transparency and lightweight deployment, and the ANN for maximum classification accuracy.

The proposed system represents a practical advancement in AIassisted petroleum reservoir engineering, with broad applicability for field screening, asset optimization, and engineering education.

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